

NARAN BULAG, China

WMO: 530830

Lat: 44.6175N

Lon: 114.1523E

Elev: 1183

StdP: 87.90

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -30.6	(c) -28.6	(d) -34.3	(e) 0.2	(f) -29.6	(g) -32.5	(h) 0.2	(i) -27.5	(j) 10.2	(k) -15.0	(l) 9.1	(m) -16.1	(n) 1.3	(o) 320	(p) 0.659

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 11.5	(c) 31.9	(d) 16.1	(e) 29.9	(f) 15.5	(g) 28.1	(h) 14.9	(i) 18.9	(j) 25.3	(k) 17.8	(l) 24.8	(m) 17.0	(n) 24.0	(o) 4.8	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 16.9	(b) 14.0	(c) 21.2	(d) 15.7	(e) 12.9	(f) 20.4	(g) 14.6	(h) 12.0	(i) 19.7	(j) 59.1	(k) 25.0	(l) 55.2	(m) 24.6	(n) 52.5	(o) 24.0	(p) 24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.5	9.9	8.6	-32.9	34.9	2.3	2.0	-34.6	36.3	-35.9	37.4	-37.2	38.5	-38.8	40.0
			-33.0	20.4	2.3	1.5	-34.7	21.5	-36.0	22.3	-37.3	23.2	-38.9	24.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.1	-20.2	-15.2	-5.4	4.6	11.9	18.2	21.3	19.2	12.8	3.1	-8.5	-17.7	
	DBStd	15.22	4.81	5.77	6.12	5.32	4.73	3.70	3.01	3.32	4.47	5.18	6.20	4.71	
	HDD10.0	3987	936	704	479	175	35	0	0	0	23	220	556	858	
	HDD18.3	6134	1194	938	737	413	204	48	7	27	172	473	806	1117	
	CDD10.0	1101	0	0	0	12	94	246	349	287	107	6	0	0	
	CDD18.3	207	0	0	0	0	5	43	98	56	6	0	0	0	
	CDH23.3	2257	0	0	0	6	129	495	979	566	82	1	0	0	
	CDH26.7	737	0	0	0	0	32	152	362	178	13	0	0	0	
Wind	WSAvg	3.6	2.5	3.0	4.1	4.8	5.0	4.0	3.6	3.4	3.6	3.7	3.1	2.6	
Precipitation	PrecAvg	222	1	2	3	6	15	35	64	63	21	8	3	2	
	PrecMax	374	4	6	14	37	51	94	133	141	52	20	9	6	
	PrecMin	125	0	0	0	0	2	4	18	5	0	0	0	0	
	PrecStd	61	1	1	3	8	12	23	28	34	14	6	3	1	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-4.6	3.1	14.4	24.0	30.4	32.7	35.0	33.9	28.5	20.4	9.1	-2.1	
		MCWB	-7.4	-2.1	4.0	8.7	12.7	15.1	17.4	17.0	13.8	9.2	2.3	-4.7	
	2%	DB	-8.1	-0.7	10.0	19.9	26.8	30.4	32.6	30.6	25.8	17.2	5.7	-6.4	
		MCWB	-10.0	-4.1	2.0	7.3	11.1	14.5	16.9	15.9	12.8	7.8	0.5	-8.2	
	5%	DB	-10.7	-3.3	7.0	17.1	24.1	28.3	30.5	28.4	23.2	14.2	2.8	-8.8	
		MCWB	-12.0	-5.8	0.1	5.9	10.3	13.9	16.4	15.3	11.5	5.8	-1.4	-10.2	
	10%	DB	-12.9	-6.2	4.1	14.3	21.7	26.2	28.5	26.4	21.0	11.5	0.2	-10.9	
		MCWB	-13.9	-8.3	-1.5	4.6	9.2	13.5	16.0	14.9	10.9	4.4	-3.0	-12.1	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-7.2	-1.5	4.5	9.7	14.1	17.5	20.9	19.8	16.5	10.6	2.9	-5.0
			MCDB	-4.9	2.2	13.0	21.8	25.5	25.3	27.2	27.3	24.4	17.6	8.7	-2.2
		2%	WB	-9.9	-3.9	2.1	7.8	12.5	16.5	19.4	18.4	14.3	8.2	0.5	-8.2
			MCDB	-8.1	-0.8	9.4	18.6	22.7	24.5	25.8	24.8	21.0	15.6	5.2	-6.4
5%		WB	-12.0	-6.0	0.3	6.4	11.3	15.6	18.4	17.4	13.1	6.5	-1.2	-10.1	
		MCDB	-10.7	-3.4	6.3	16.2	21.1	23.8	25.4	23.7	20.5	13.2	2.4	-8.8	
10%		WB	-13.9	-8.3	-1.3	4.9	10.0	14.8	17.6	16.4	12.0	4.8	-2.8	-12.0	
		MCDB	-13.0	-6.3	3.8	13.9	19.9	23.2	24.9	23.2	19.1	10.8	0.3	-10.9	
Mean Daily Temperature Range		5% DB	MDBR	10.0	11.4	12.3	13.1	13.5	12.0	11.5	11.9	12.5	12.1	11.0	9.6
			MCDBR	11.9	13.6	15.4	17.1	17.9	15.3	14.6	14.8	15.2	15.6	13.5	11.4
	MCWBR		10.4	10.6	9.5	8.8	8.2	5.3	4.5	5.0	6.5	8.4	9.2	9.8	
	5% WB	MCDBR	11.8	13.3	14.9	15.9	15.0	11.9	10.9	11.4	12.6	14.5	12.9	11.2	
		MCWBR	10.4	10.6	9.5	8.5	7.4	4.4	4.6	4.9	6.1	8.4	9.1	9.7	
	Clear-Sky Solar Irradiance	taub	0.256	0.287	0.328	0.379	0.437	0.459	0.420	0.404	0.359	0.320	0.279	0.259	
taud		2.323	2.245	2.178	2.085	1.964	2.014	2.198	2.205	2.266	2.304	2.312	2.292		
Ebn at Noon		873	899	901	878	835	816	846	846	857	846	829	824		
Edh at Noon		78	102	127	152	178	170	140	134	116	96	78	73		
All-Sky Solar Radiation	RadAvg	2.10	3.23	4.78	6.04	6.69	6.52	6.41	5.94	4.92	3.57	2.34	1.73		
	RadStd	0.22	0.24	0.29	0.31	0.40	0.42	0.24	0.21	0.25	0.21	0.18	0.18		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (4 neighbors)	N/A	N/A	-1.71	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	-1.54	N/A	N/A	-0.65	N/A	N/A	N/A	N/A

Nomenclature: See separate page